

ProductionManagementAI — Identity Schema — Database Design Document (テーブル定義書)

000_DB — supports auth/RBAC foundation (`work-items/WI-001/brief.md` INFRA-002), implements 0002_ADR (`docs/en/architecture/0002/0002_ADR_auth-rbac-foundation.md`).

Physical naming convention: `snake_case` , per `ai/skills/database-design/SKILL.md` ("until [a convention is] fixed, use `snake_case` physical names consistently"). ASP.NET Core Identity's default PascalCase table names (`AspNetUsers` , etc.) are remapped to `snake_case` via EF Core Fluent API `.ToTable(...)` calls in `AppDbContext` , and `EFCore.NamingConventions` (proposed in `work-items/WI-001/plan.md` , Auth foundation design) `snake_cases` column names consistently.

Table list

Table	Physical name	Purpose
Users	<code>users</code>	Application user accounts (Identity <code>IdentityUser<Guid></code> + app-specific fields)
Roles	<code>roles</code>	RBAC roles (Identity <code>IdentityRole<Guid></code>); seeded <code>Admin</code> , <code>Operator</code>
UserRoles	<code>user_roles</code>	Many-to-many join between users and roles
UserClaims	<code>user_claims</code>	Identity default — per-user claims (unused by any feature yet, kept for extensibility)
RoleClaims	<code>role_claims</code>	Identity default — per-role claims (unused by any feature yet, kept for extensibility)
UserLogins	<code>user_logins</code>	Identity default — external login providers (unused; no external login in scope)
UserTokens	<code>user_tokens</code>	Identity default — per-user tokens, e.g. password-reset (unused by any feature yet, kept for extensibility)

ER diagram and relationships

Table	Related table	Relationship (1:1 / 1:N / N:N)	FK column
<code>users</code>	<code>roles</code>	N:N (via <code>user_roles</code>)	<code>user_roles.user_id</code> / <code>user_roles.role_id</code>
<code>users</code>	<code>user_claims</code>	1:N	<code>user_claims.user_id</code>

Table	Related table	Relationship (1:1 / 1:N / N:N)	FK column
users	user_logins	1:N	user_logins.user_id
users	user_tokens	1:N	user_tokens.user_id
roles	role_claims	1:N	role_claims.role_id

Table definitions

users

Item name	Physical name	Data type	Length	NOT NULL	Default	PK/FK	Notes
Id	id	uuid	—	yes	gen_random_uuid()	PK	Identity IdentityUser<Guid> key
UserName	user_name	varchar	256	no	none	—	Identity default
NormalizedUserName	normalized_user_name	varchar	256	no	none	—	Uppercased for case-insensitive lookup; unique-indexed
Email	email	varchar	256	no	none	—	Identity default
NormalizedEmail	normalized_email	varchar	256	no	none	—	Uppercased for case-insensitive lookup
EmailConfirmed	email_confirmed	boolean	—	yes	false	—	Identity default; not used by any feature yet (no email-confirmation flow in scope)
PasswordHash	password_hash	text	—	no	none	—	Identity-managed hash; never read/written outside Identity APIs
SecurityStamp	security_stamp	text	—	no	none	—	Identity default; invalidated on credential change
ConcurrencyStamp	concurrency_stamp	text	—	no	none	—	Identity default; optimistic concurrency
PhoneNumber	phone_number	varchar	32	no	none	—	Identity default; unused
PhoneNumberConfirmed	phone_number_confirmed	boolean	—	yes	false	—	Identity default; unused
TwoFactorEnabled	two_factor_enabled	boolean	—	yes	false	—	Identity default; no 2FA flow in scope
LockoutEnd	lockout_end	timestampz	—	no	none	—	Identity default; supports STRIDE lockout mitigation in 0002_ADR
LockoutEnabled	lockout_enabled	boolean	—	yes	true	—	Identity default
AccessFailedCount	access_failed_count	integer	—	yes	0	—	Identity default

Item name	Physical name	Data type	Length	NOT NULL	Default	PK/FK	Notes
DisplayName	display_name	varchar	200	yes	none	—	App-specific addition
IsActive	is_active	boolean	—	yes	true	—	App-specific addition; reserved for future account-disable use case, not yet consumed by any requirement
CreatedAtUtc	created_at_utc	timestampz	—	yes	now()	—	App-specific addition

roles

Item name	Physical name	Data type	Length	NOT NULL	Default	PK/FK	Notes
Id	id	uuid	—	yes	gen_random_uuid()	PK	Identity IdentityRole<Guid> key
Name	name	varchar	256	no	none	—	Admin / Operator , seeded
NormalizedName	normalized_name	varchar	256	no	none	—	Uppercased; unique-indexed
ConcurrencyStamp	concurrency_stamp	text	—	no	none	—	Identity default

user_roles

Item name	Physical name	Data type	Length	NOT NULL	Default	PK/FK	Notes
UserId	user_id	uuid	—	yes	none	PK, FK → users.id	Composite PK with role_id
RoleId	role_id	uuid	—	yes	none	PK, FK → roles.id	Composite PK with user_id

user_claims , role_claims , user_logins , user_tokens

Kept at ASP.NET Core Identity's standard EF Core column mapping (near-zero cost to retain, per the Auth foundation design section of `work-items/WI-001/plan.md` : "avoids a breaking schema change if external login/claims are needed later"). Not itemized column-by-column here since no column deviates from Identity's default shape; generated automatically by `IdentityDbContext<AppUser, AppRole, Guid, ...>` and reviewed at migration-generation time (`work-items/WI-001/plan.md` step 11).

Index definitions

Index name	Table	Column(s)	Type	Rationale (access pattern)
<code>ix_users_normalized_user_name</code>	<code>users</code>	<code>normalized_user_name</code>	unique btree	Login looks up by normalized username; Identity requires uniqueness
<code>ix_users_normalized_email</code>	<code>users</code>	<code>normalized_email</code>	btree	Supports future email-based lookup; not unique (Identity's default <code>RequireUniqueEmail</code> is off unless explicitly enabled — left off since brief.md doesn't require email-based login)
<code>ix_roles_normalized_name</code>	<code>roles</code>	<code>normalized_name</code>	unique btree	Role lookup/seed-idempotency by name
<code>pk_user_roles</code>	<code>user_roles</code>	<code>user_id, role_id</code>	unique btree (PK)	Primary lookup path: "what roles does this user have"

Constraints

Constraint	Type (PK/FK/UNIQUE/CHECK)	Table.column(s)	References	Rule
<code>pk_users</code>	PK	<code>users.id</code>	—	—
<code>pk_roles</code>	PK	<code>roles.id</code>	—	—
<code>pk_user_roles</code>	PK	<code>user_roles.(user_id, role_id)</code>	—	—
<code>fk_user_roles_user_id</code>	FK	<code>user_roles.user_id</code>	<code>users.id</code>	<code>ON DELETE CASCADE</code> — removing a user removes their role assignments
<code>fk_user_roles_role_id</code>	FK	<code>user_roles.role_id</code>	<code>roles.id</code>	<code>ON DELETE CASCADE</code> — removing a role removes its assignments (role deletion itself is not exposed by any feature yet)
<code>uq_users_normalized_user_name</code>	UNIQUE	<code>users.normalized_user_name</code>	—	Enforced by <code>ix_users_normalized_user_name</code>
<code>uq_roles_normalized_name</code>	UNIQUE	<code>roles.normalized_name</code>	—	Enforced by <code>ix_roles_normalized_name</code>

API / DD mapping

Table.column	DD field	API field
users.id	not yet written (WI-001 has no DD; auth endpoints are specified directly in 0002_ADR)	GET /api/auth/me → id
users.user_name	—	GET /api/auth/me → userName ; POST /api/auth/login request → userName
users.display_name	—	GET /api/auth/me → displayName
roles.name (via user_roles)	—	GET /api/auth/me → roles: string[]

Migration impact and recovery limits

- **Migration type:** additive — this is the first migration in the project; no existing data to preserve or transform.
- **Data recovery limit:** not applicable — no pre-existing rows in any of these tables.
- **Rollback plan:** `dotnet ef database update <previous-migration-or-0>` drops these tables cleanly since nothing else references them yet; safe to roll back at this stage.

Open decisions

Decision	Options	Recommendation	Status
Whether an additional CSRF token (beyond SameSite cookie attribute) is needed for state-changing endpoints	SameSite=Lax only vs. SameSite=Lax + anti-forgery token	SameSite=Lax only — no additional token	decided — work-items/WI-002/decisions.md DEC-020 (2026-09-18)
Whether email_confirmed / two_factor_enabled / phone_number* columns are ever exercised by a future requirement	keep unused vs. trim from the model	Keep (near-zero cost, avoids a later breaking migration) — already decided, see users table Notes	decided — see work-items/WI-001/decisions.md (auth foundation design)